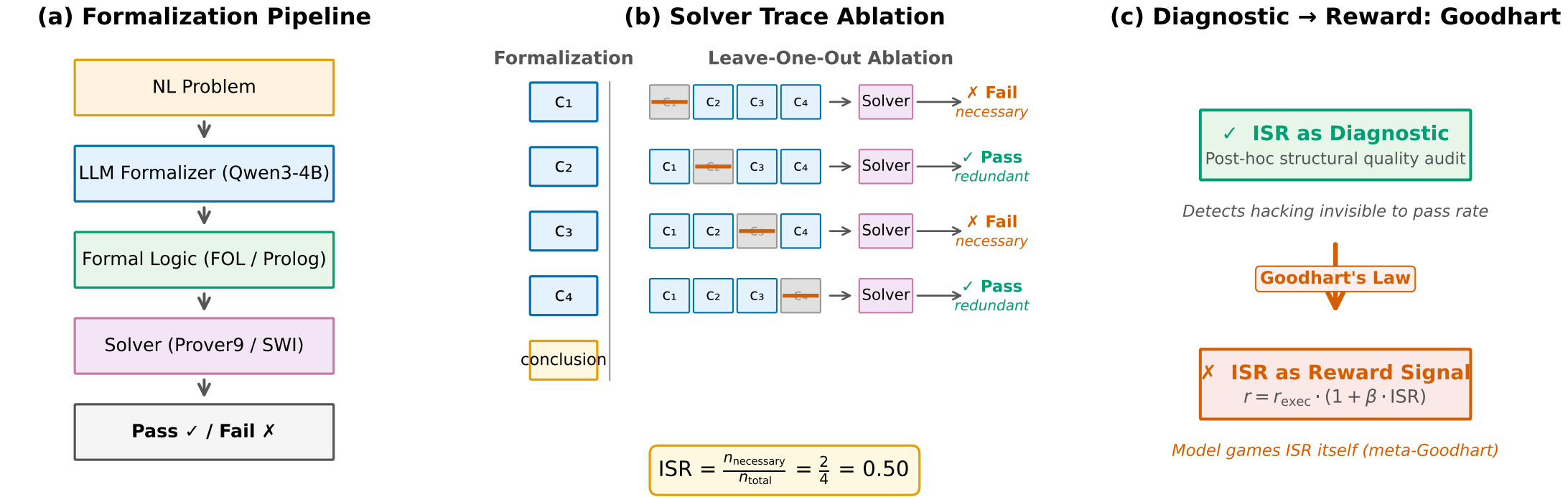
(a) Formalization Pipeline
NL Problem
LLM Formalizer (Qwen3-4B)
Formal Logic (FOL / Prolog)
Solver (Prover9 / SWI)
Pass ✓ / Fail ✗
(b) Solver Trace Ablation
Formalization
Leave-One-Out Ablation
C1
C2
C3
C4
conclusion
Solver
✗ Fail
necessary
✓ Pass
redundant
ISR = n_necessary / n_total = 2/4 = 0.50
(c) Diagnostic → Reward: Goodhart
✓ ISR as Diagnostic
Post-hoc structural quality audit
Detects hacking invisible to pass rate
Goodhart's Law
✗ ISR as Reward Signal
r = r_exec · (1 + β · ISR)
Model games ISR itself (meta-Goodhart)